Physics-Informed Multimodal Predictive Maintenance and Digital Twin of Data-Center Cooling Equipment

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***Abstract*—Degradation of chilled-water pumps and computer-room air-conditioning (CRAC) fans lowers cooling capacity gradually, and threshold alarms on server inlet temperature fire only after the thermal margin has been consumed. This paper presents a simulation-based predictive-maintenance framework for a pump and a fan in which every fault is parameterized by a single severity variable $D \in [0, 1]$. Reduced-order hydraulic and induction-motor models feed synthetic vibration and motor-current generators; the resulting signals are conditioned, converted to time- and frequency-domain features, fused with operating-state variables, and classified. A Health Index derived from the worst active severity, a trend-based remaining-useful-life (RUL) estimate, and a lumped first-order thermal model of the server zone are coupled through an explicit fault-to-cooling-effectiveness map. A rule-based decision engine applies fixed thresholds on Health Index, RUL, and thermal margin and returns the most urgent maintenance level. Fault injection, threshold definition, and the exact-exponential thermal update used to remain stable at hour-scale time steps are described in detail. A predictive policy is compared with reactive and no-intervention policies over identical degradation trajectories, and the early-warning lead time is measured rather than assumed; the measured lead time was [lead time, h]. All results derive from synthetic data, and no experimental validation is claimed.**

***Index Terms*—Predictive maintenance, digital twin, fault diagnosis, motor current signature analysis, remaining useful life, data-center cooling.**

I. Introduction

Cooling accounts for a large share of the electrical load of a data center, and it is delivered by rotating machinery: chilled-water pumps move heat from the room to the plant, and CRAC fans move air across the heat exchangers [1], [2]. The thermal envelope that this machinery must hold is specified by operating guidelines for the inlet air of IT equipment [3]. When a pump cavitates or a fan bearing wears, flow and heat-transfer capacity fall, and the server inlet temperature rises toward the limit of that envelope.

Condition-based maintenance replaces fixed-interval servicing with decisions driven by measured machine state [4], and prognostics and health management (PHM) extends it with estimates of how long a degrading asset can continue to operate [5]. Industrial practice for vibration monitoring and general machine diagnostics is codified in ISO 13373-1 and ISO 17359 [6], [7]. These documents define procedures and alarm concepts, but they do not link a machine fault to its consequence for the process the machine serves.

Three gaps motivate this work. First, vibration-only and current-only diagnosis each miss fault classes that the other modality exposes, so a fused feature set is needed. Second, health indicators are usually reported per machine and are not translated into server temperature, which is the quantity that operators must protect. Third, the benefit of predictive over reactive maintenance is often asserted rather than measured on a common trajectory. Physical run-to-failure data for cooling pumps and fans are scarce, so the work is carried out entirely in simulation, and its claims are limited accordingly.

The contributions are as follows.

1) A severity-parameterized fault model in which every fault of a chilled-water pump and a CRAC fan acts through one variable D, with explicit signal signatures and efficiency couplings.
2) A multimodal diagnosis chain that fuses vibration features, motor-current-signature features, and operating-state features, evaluated with scenario-level data splits and a five-step ablation ladder.
3) A Health Index and a trend-based RUL estimator that returns no value, rather than an invented one, when the trend does not degrade.
4) A fault-to-thermal link that converts equipment degradation into cooling effectiveness and into server inlet temperature through an exactly integrated first-order model.
5) A documented threshold set and decision engine, and a controlled comparison of no-intervention, reactive, and predictive policies with a measured early-warning lead time.

The remainder of the paper is organized as follows. Section II reviews related work. Section III describes the models, fault injection, and thresholds. Section IV gives the experimental protocol and results. Section V concludes.

II. Related Work

A. Vibration and Current Based Diagnosis

Rolling-element bearing diagnosis from vibration relies on characteristic defect frequencies and on envelope analysis

of the resonance excited by repetitive impacts [8]. Public benchmark data have been used to compare such methods under a common protocol [9]. Motor current signature analysis (MCSA) detects mechanical and electrical faults through sidebands in the stator current spectrum without a dedicated vibration sensor [10], [11], and the underlying machine behavior is treated in depth in [12].

Machine-learning methods map extracted or learned features to fault classes. Broad surveys cover feature-based classifiers and deep networks [13], [14], with bearing diagnosis as the most studied case [15], [16]. Ensemble trees remain a robust baseline for tabular features [17], while temporal convolutional networks have been proposed as a general sequence-modeling alternative to recurrent networks [18]. Model-based diagnosis, based on residuals between measured and modeled behavior, offers a complementary route [19], [20], and data-driven process monitoring has been reviewed in [21].

B. Prognostics

Reviews of machinery prognostics cover the path from data acquisition to RUL prediction [22] and the statistical approaches to RUL estimation [23]. Vibration-derived degradation signals have been mapped to residual life with neural networks [24], and run-to-failure simulation datasets such as the turbofan benchmark of Saxena *et al.* [25] have standardized comparison. These studies treat the machine in isolation; the consequence of the predicted failure for the cooled process is outside their scope.

C. Digital Twins and Physics-Informed Learning

The digital-twin concept couples a virtual representation to a physical asset over its life cycle [26], and its industrial use has been surveyed in [27]. Physics-informed learning embeds governing equations or physical residuals in data-driven models [28], [29], and hybrid physics–data models have been discussed for manufacturing systems [30]. The present work follows this direction in a restrained form: physics supplies the signal generators, the efficiency couplings, and the thermal model, and a compact set of physics-derived residual features is tested in an ablation rather than assumed to help.

III. Methodology and System Architecture

A. Overview

Fig. 1 shows the processing chain. One call of the engine evaluates one operating point and returns the injected fault, the predicted fault, the Health Index, the RUL, the cooling effectiveness, the server inlet temperature, the thermal margin, and the maintenance level. Vibration and current signals are regenerated for the current operating point at every step, since a sensor reports only the present state. The Health Index, the RUL, and the thermal trajectory are computed over an accumulated history buffer, because the RUL fit and the thermal integration both require a time series.

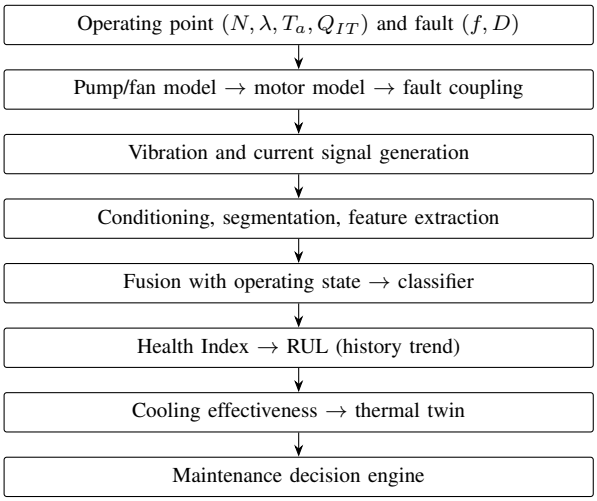


Fig. 1. Signal flow of the digital twin. Each block corresponds to one implementation module.

B. Hydraulic and Motor Models

Pump and fan performance is scaled from the nominal operating point with the affinity laws,

$$Q \propto N, \qquad \Delta P \propto N^2, \qquad P_{\mathrm{sh}} \propto N^3, \tag{1}$$

where N is shaft speed. Efficiency follows a quadratic curve about the best-efficiency point,

$$\eta(\lambda) = \eta_{\mathrm{nom}} \left[1 - k(\lambda - 1)^2\right], \quad k = 0.30, \tag{2}$$

with λ the load fraction. The drive is an induction motor with synchronous speed $N_s = 120 f_\ell / p$ and slip $s = (N_s - N)/N_s$, where f_ℓ is the line frequency and p the pole count. Electrical power and line current follow from

$$P_{\mathrm{el}} = \frac{\tau \omega}{\eta_m}, \qquad I = \frac{P_{\mathrm{el}}}{\sqrt{3}\, V\, \mathrm{pf}}, \tag{3}$$

with constant $\eta_m = 0.90$ and $\mathrm{pf} = 0.85$. These are reduced-order surrogates and are not validated against CFD or measurements.

C. Fault Injection

1) Severity variable and applicable faults: Each fault is injected through a single severity $D \in [0, 1]$, where $D = 0$ is the healthy state and $D = 1$ is the end-of-life state of that fault. The Healthy label requires $D = 0$; a request with $D > 0$ for a Healthy label is rejected by the input validator. The label set depends on the asset. The pump admits Healthy, Cavitation, Unbalance, BearingDegradation, and Misalignment, and the fan admits Healthy, Unbalance, BearingDegradation, Misalignment, and EfficiencyDegradation. A label that does not belong to the selected asset is rejected rather than silently remapped. The injected operating conditions are drawn within fixed ranges (Table I).

TABLE I
OPERATING RANGES USED FOR SCENARIO GENERATION AND CONTROL

Variable	Range	Unit
Shaft speed N	1200–1800	rpm
Load fraction λ	0.3–1.0	–
Ambient temperature T_a	18–32	°C
IT heat load Q_{IT}	10–100	kW
Fault severity D	0–1	–

2) Fault-to-efficiency coupling: A fault reduces hydraulic performance through

$$\eta_f = \eta_{\text{nom}}\big(1 - g_f(D)\big), \tag{4}$$

where g_f is monotone with $g_f(0) = 0$, and fault-specific coefficients scale flow and head in the same way. Cavitation is driven by a cavitation severity C derived from the net positive suction head margin $\Delta_{\text{NPSH}} = \text{NPSH}_a - \text{NPSH}_r$; C, not Δ_{NPSH}, controls flow, head, and efficiency loss and the broadband vibration energy. This is a surrogate and not a cavitation model.

3) Signal signatures: Table II lists the signature that each fault leaves in the sensor signals. Bearing defect frequencies follow from the bearing geometry, with N_b rolling elements, ball diameter B_d, pitch diameter P_d, contact angle ϕ, and shaft frequency f_r:

$$\text{BPFO} = \tfrac{N_b}{2} f_r \Big(1 - \tfrac{B_d}{P_d}\cos\phi\Big), \tag{5}$$

$$\text{BPFI} = \tfrac{N_b}{2} f_r \Big(1 + \tfrac{B_d}{P_d}\cos\phi\Big), \tag{6}$$

$$\text{BSF} = \tfrac{P_d}{2B_d} f_r \Big[1 - \big(\tfrac{B_d}{P_d}\cos\phi\big)^2\Big], \tag{7}$$

$$\text{FTF} = \tfrac{f_r}{2}\Big(1 - \tfrac{B_d}{P_d}\cos\phi\Big). \tag{8}$$

The vibration signal is synthesized as

$$\begin{aligned} x(t) &= A_0 \sin(2\pi f_r t) + \sum_{h\geq 2} A_h \sin(2\pi h f_r t) \\ &+ a_u(U)\sin(2\pi f_r t) + a_m(M)\big[\sin(2\pi f_r t) + \sin(4\pi f_r t)\big] \\ &+ a_b(D_b)\sum_k \psi(t - kT_b) + a_c(C)\, n_{\text{bb}}(t) + n_s(t), \end{aligned} \tag{9}$$

where U, M, and D_b are the unbalance, misalignment, and bearing severities, $T_b = 1/\text{BPFO}$, ψ is an exponentially decaying pulse, n_{bb} is band-limited broadband noise, and n_s is sensor noise. The unbalance amplitude scales as $F_u = me\omega^2$. The stator current is

$$i(t) = I_0\big[1 + m_d \sin(2\pi f_r t)\big]\sin(2\pi f_\ell t) + \text{harmonics} + n_i(t), \tag{10}$$

with modulation depth $m_d \propto U + M + D_b$, which reproduces the sidebands at $f_\ell \pm f_r$ produced when a periodic mechanical disturbance modulates torque and slip [10], [11].

D. Signal Conditioning and Features

Vibration and current records are conditioned (filtering and normalization) and segmented into windows. Vibration features include time-domain statistics, band energies, and the spectral amplitudes at f_r, $2f_r$, and the bearing frequencies of (5)–(8). Current features include the RMS value and the sideband amplitudes around f_ℓ. Operating-state features comprise speed, load fraction, ambient temperature, and motor-derived electrical power. A compact set of four physics-residual features, each the difference between a measured quantity and its model prediction, is evaluated separately in the ablation.

TABLE II
FAULT CLASSES, SIGNAL SIGNATURES, AND THERMAL CONSEQUENCE

Fault	Asset	Signature	Thermal penalty
Cavitation	Pump	Broadband high-frequency vibration; flow, head, efficiency loss	Largest
Unbalance	Both	1X vibration; current modulation	Small residual
Misalignment	Both	1X and 2X vibration; current modulation	Small residual
Bearing degradation	Both	Impulse train at BPFO; current modulation via D_b	Small residual
Efficiency degradation	Fan	Little vibration or current change; efficiency loss	Airflow-related

E. Classification

A random forest [17] maps the fused feature vector to a fault label. When the statistics toolbox is not available, a k-nearest-neighbor classifier is used as a documented fallback. The classifier returns labels only; it produces no confidence value and no severity estimate, and none is displayed. The classifier sees extracted features exclusively, so the injected label never reaches it. All train/test splits are made at scenario level, so that windows from one scenario cannot appear on both sides of a split.

F. Health Index and Remaining Useful Life

The Health Index is

$$\text{HI} = 100\left(1 - \min\left\{\frac{D_{\text{eff}}}{D_{\text{crit}}}, 1\right\}\right), \qquad D_{\text{eff}} = \max_j D_j, \tag{11}$$

where D_j ranges over the active faults of the asset and $D_{\text{crit}} =$ **[D_crit from config]** is the severity at which health reaches zero. For the RUL, a straight line $\text{HI}(t) = a + bt$ is fitted by least squares over a recent window of n history points. With t_k the current time and HI_c the critical Health Index,

$$\widehat{\text{RUL}} = \begin{cases} \dfrac{\text{HI}_c - a}{b} - t_k, & b < 0, \\ \text{undefined}, & b \geq 0. \end{cases} \tag{12}$$

When the slope is non-negative or fewer than two points exist, no RUL is reported. The number of points behind each estimate is exposed, and an estimate resting on fewer than five points is labeled provisional.

G. Fault-to-Thermal Link and Thermal Twin

Degradation reduces the effective cooling capacity through a cooling effectiveness $\varepsilon \in (0, 1]$,

$$\varepsilon_k = 1 - h_f(D_k), \tag{13}$$

with h_f monotone and $h_f(0) = 0$. Pump cavitation and the fan faults that reduce airflow receive the largest penalty; purely mechanical faults receive a smaller residual penalty (Table II). The server zone is a lumped first-order system with thermal capacitance C_{th} and time constant τ,

$$C_{th}\frac{dT}{dt} = Q_{IT} - \varepsilon\,\frac{C_{th}}{\tau}(T - T_a). \tag{14}$$

The simulation advances in steps of Δt on the order of one hour, while τ is on the order of seconds. Forward Euler is unstable at that ratio and diverged to oscillating values of very large magnitude in this project. Each interval is therefore integrated in closed form,

$$T_{ss,k} = T_a + \frac{Q_{IT}\,\tau}{\varepsilon_k C_{th}}, \tag{15}$$

$$T_{k+1} = T_{ss,k} + \big(T_k - T_{ss,k}\big)\exp\!\Big(-\frac{\varepsilon_k \Delta t}{\tau}\Big), \tag{16}$$

which is stable for any $\Delta t > 0$. The thermal margin is $M_k = T_{\text{safe}} - T_k$, with $T_{\text{safe}} =$ **[T_safe, °C]**. Each trajectory starts at ambient temperature, so the first reported temperature equals T_a and the settled value appears from the second step.

H. Thresholds and Maintenance Decision

The decision engine evaluates three signals, Health Index, RUL, and thermal margin, against fixed thresholds (Table III). The thresholds were set once and are not tuned per experiment. Through (11), the Health Index thresholds correspond to fixed fractions of the critical severity:

$$\frac{D_{\text{eff}}}{D_{\text{crit}}} = 1 - \frac{\text{HI}_{\text{thr}}}{100} \;\Rightarrow\; 0.30,\ 0.60,\ 0.85, \tag{17}$$

for $\text{HI}_{\text{thr}} = 70, 40, 15$. A fault is therefore flagged for warning once it reaches 30% of the critical severity, for maintenance at 60%, and as critical at 85%. Each signal is mapped to one of five levels (NORMAL, MONITOR, WARNING, MAINTENANCE_REQUIRED, CRITICAL), and the engine returns the most urgent level over the three signals,

$$L = \max\{L_{\text{HI}},\, L_{\text{RUL}},\, L_M\}. \tag{18}$$

The diagnostic label is not used as a threshold input; it is reported alongside the level so that disagreement between injected and predicted faults remains visible.

I. Single-Step Procedure

Algorithm 1 summarizes one evaluation of the engine.

TABLE III
MAINTENANCE DECISION THRESHOLDS

Signal	Threshold	Value
Health Index	Warning	70
Health Index	Maintenance	40
Health Index	Critical	15
RUL	Warning	24 h
RUL	Maintenance	8 h
Thermal margin	Warning	5 °C
Thermal margin	Critical	1 °C

Algorithm 1 One Simulation Step

Require: asset e, fault f, severity D, $(N, \lambda, T_a, Q_{IT})$, history $\mathcal{H}$, trained model $\mathcal{M}$

1: Validate $(e, f, D, N, \lambda, T_a, Q_{IT})$ against allowed labels and ranges; reject otherwise
2: Evaluate hydraulic and motor models (1)–(3); apply fault coupling (4)
3: Generate $x(t)$ by (9) and $i(t)$ by (10); condition and window both
4: Extract feature vector $\mathbf{z}$ and fuse with operating state
5: $\hat{f} \leftarrow \mathcal{M}(\mathbf{z})$
6: Compute HI by (11); append $(t, \text{HI}, \varepsilon)$ to $\mathcal{H}$
7: Estimate RUL by (12) over $\mathcal{H}$
8: Compute ε by (13); update T by (15)–(16); compute M
9: $L \leftarrow$ decision engine (18)
10: **return** $(f, \hat{f}, D, \text{HI}, \widehat{\text{RUL}}, \varepsilon, T, M, L)$

J. Interactive Deployment Layer

The engine is exposed to an operator interface through a Gateway of an industrial supervisory platform. The MATLAB process acts only as a client: it polls a command endpoint, runs Algorithm 1, and posts telemetry to a second fixed endpoint. Both endpoints require a shared key and a loopback client address. Commands are validated against the label sets and ranges above before they reach the engine, and telemetry is range- and label-checked again before any tag is written. The interface displays the injected and the predicted fault in separate fields, shows no RUL when none exists, and marks the backend as disconnected when telemetry is older than a configurable interval. Nothing in this layer drives physical equipment.

IV. Experimental Setup and Results

A. Setup

All data were produced by the simulator of Section III. Scenarios were generated with the random seed fixed in the project configuration, with N, λ, T_a, and Q_{IT} drawn within Table I and D spanning $[0, 1]$. Train and test sets were separated at scenario level, and the leakage audit, parameter sanity checks, fault-injection monotonicity checks, and reproducibility checks were run by the validation script. The dataset contained **[number of scenarios]** scenarios and **[number of windows]** windows, split **[train/test ratio]**.

TABLE IV
DIAGNOSIS ABLATION (SCENARIO-LEVEL TEST SPLIT)

Feature set	Accuracy (%)	Macro-F1
Vibration only	[]	[]
Current only	[]	[]
Vibration + current	[]	[]
+ operating state	[]	[]
+ physics residuals	[]	[]

TABLE V
LEVEL CROSSINGS DURING A SEVERITY RAMP

Level reached	Time (h)	RUL error (h)
WARNING	[]	[]
MAINTENANCE_REQUIRED	[]	[]
CRITICAL	[]	[]

TABLE VI
MAINTENANCE POLICY COMPARISON

Policy	$T_{\max}$ (°C)	t_{viol} (h)	Intervention (h)
None	[]	[]	–
Reactive	[]	[]	[]
Predictive	[]	[]	[]

TABLE VII
THRESHOLD SENSITIVITY

Perturbed threshold	Δt_{lead} (h)	False escalations (%)
$\text{HI}_{\text{warning}}$	[]	[]
$\text{HI}_{\text{maintenance}}$	[]	[]
RUL maintenance	[]	[]
Thermal margin warning	[]	[]

B. Metrics

Diagnosis was scored by accuracy and macro-averaged F1 over the fault classes. Thermal performance was scored by the peak server inlet temperature $T_{\max}$ and by the time t_{viol} spent with $M < 0$. The early-warning lead time is

$$\Delta t_{\text{lead}} = t_{\text{react}} - t_{\text{pred}}, \tag{19}$$

where t_{react} is the time at which the reactive policy first intervenes, that is, after the safe temperature has been violated, and t_{pred} is the time at which the predictive policy first intervenes on the basis of Health Index, RUL, or thermal-margin thresholds.

C. Diagnosis and Ablation

Five feature sets were compared: vibration only, current only, vibration and current fused, fusion with operating-state features, and fusion with operating state and physics-residual features. Table IV reports the results.

A known case should be stated. In an earlier backend test, a healthy fan was labeled EfficiencyDegradation. That fault leaves almost no vibration or current signature, and the classifier receives no flow, pressure, or power measurement, so the confusion follows from the sensing set and was left uncorrected. The per-class confusion matrix **[Fig. X]** should be examined for the pairs Healthy/EfficiencyDegradation and Unbalance/Misalignment.

D. Health Index and RUL Behavior

Severity was ramped from 0 to 1 in hourly steps for the fan bearing and for pump cavitation. Table V lists the Health Index at which each decision level was first reached and the RUL error at those instants. The RUL is absent until at least two points with a falling Health Index exist, and it is provisional below five points.

E. Reactive Versus Predictive Maintenance

Three policies were run on identical severe-degradation trajectories: no intervention, reactive (intervene after the thermal limit is violated), and predictive (intervene when the decision level reaches the maintenance threshold). Table VI lists the outcome. The values are those printed by the experiment script and may change slightly with the random scenario draw.

The measured lead time of (19) was **[Δt_{lead}, h]**. A fault with a small thermal penalty, such as bearing degradation under Table II, yields a small temperature effect in this model, and no compensation was added to make the case more pronounced.

F. Threshold Sensitivity

Each threshold in Table III was perturbed by $\pm$**[x]**% to determine how Δt_{lead} and the false-escalation rate on Healthy trajectories change. Results are in Table VII.

G. Limitations

The physics models are reduced-order and have not been checked against CFD or measurements. The cavitation and thermal models are simplifications. The Health Index and RUL are analytic baselines, and a learned Health Index or a sequence-model RUL, for instance a temporal convolutional network [18], has not been built. The physics-residual set contains four features. All metrics come from synthetic data, and no experimental validation is claimed. The interactive layer in Section III-I had not been executed against a running Gateway at the time of writing.

V. CONCLUSION AND FUTURE WORK

A simulated pump and CRAC fan were modeled with severity-parameterized faults, and the chain from fault injection to sensor signature, diagnosis, Health Index, RUL, server inlet temperature, and maintenance level was implemented with fixed, documented thresholds. A predictive policy was compared with reactive and no-intervention policies on common trajectories, and the early-warning lead time was measured. The framework is a basis for later work that replaces the synthetic generators with recorded signals at the interface where simulated vibration and current enter the pipeline.

Planned extensions are a learned Health Index compared with the analytic form, a sequence-model RUL estimator, a larger physics-residual set, additional assets beyond one pump and one fan, and validation on run-to-failure data from physical equipment.

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